

Megnath Ramesh

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Education

- 2020–2025 **Ph.D. in Electrical and Computer Engineering**, *University of Waterloo*, Waterloo, ON
Thesis: Planning and Replanning Near-optimal Robot Coverage Paths in Partially Unknown Environments.
Advisors: Stephen L. Smith and Baris Fidan
Key Courses: Combinatorial Optimization, Computational Geometry, Deep Learning, Algorithm Design.
- 2013–2018 **B.Sc. in Electrical Engineering**, *University of Alberta*, Edmonton, AB
GPA: 3.8/4.0 - Graduated with Distinction.

Research Appointments

- 2025–Present **Postdoctoral Fellow (Vector Faculty Affiliate Researcher)**, *University of Waterloo*, Waterloo, ON
Advisors: Stephen L. Smith and Baris Fidan
Developing generative AI (diffusion) models to plan robot paths for long-horizon coverage and exploration tasks in unknown environments.
- 2020–2025 **Ph.D. Researcher**, *University of Waterloo & Avidbots Corp.*, Waterloo, ON
- Developed efficient coverage planning algorithms for industrial cleaning robots.
 - Tested algorithms using real-world robots that encounter environmental uncertainties.
 - Collaborated with the Avidbots engineering team on algorithm implementation and testing.
 - Mentored junior researchers in devising and implementing solutions to difficult robotics problems.

Publications

BenchNPIN - Benchmarking Non-Prehensile Interactive Navigation.

N. Zhong, S. Caro, A. Iskander, **M. Ramesh**, and S. L. Smith.

Under review for potential publication at an IEEE conference.

Minimum-Length Coverage Path Planning for Grid Environments with Approximation Guarantees.

M. Ramesh, F. Imeson, B. Fidan, and S. L. Smith.

IEEE Robotics and Automation Letters (RA-L), 2025.

Approximate Environment Decompositions for Robot Coverage Planning using Submodular Set Cover.

M. Ramesh, F. Imeson, B. Fidan, and S. L. Smith.

IEEE Conference on Decision and Control (CDC), 2024.

Anytime Replanning of Robot Coverage Paths for Partially Unknown Environments.

M. Ramesh, F. Imeson, B. Fidan, and S. L. Smith.

IEEE Transactions on Robotics (T-RO), 2024.

Optimal Partitioning of Non-Convex Environments for Minimum Turn Coverage Planning.

M. Ramesh, F. Imeson, B. Fidan, and S. L. Smith.

IEEE Robotics and Automation Letters (RA-L), 2022.

Teaching Experience

- 2021–2025 **Graduate Teaching Assistant**, *University of Waterloo*, Waterloo, ON
- Courses:**
- ECE 406: Algorithm Design and Analysis (Winter 2024 & Winter 2025)
 - MTE 484: Digital Control Applications (Fall 2024)
 - ECE 484: Digital Control Applications (Fall 2023)
 - ECE 380: Analog Control Systems (Spring 2021)

Industry Experience

- 2018–2020 **Application Software Developer (L1 & L2)**, *ScopeAR*, Edmonton, AB
- Developed features and fixed defects in a cross-platform augmented reality (AR) video calling application.
 - Worked on a range of features, including marker-based AR tracking, guest login, and cloud-based AR.
- 2017–2018 **Junior Robotics Developer**, *VEERUM*, Calgary, AB
- Developed a pilot project using autonomous robots for generating digital twins of construction facilities.
- May – August 2016 **Control System Software Developer (Co-op)**, *University of Alberta*, Edmonton, AB
- Developed an application to safely control and visually inspect the integrity of a dual thick-wall pipe in a high-pressure pumping system.
- January – April 2016 **Robotics Research Intern (Co-op)**, *École Polytechnique de Montréal*, Montreal, QC
- Integrated and tested an autonomous mapping algorithm using a Clearpath Husky with a depth camera mounted on a robotic arm.

Notable Honors and Awards

- 2023–2025 2x Ontario Graduate Scholarship - University of Waterloo
- 2021–2025 3x UW President's Graduate Scholarship - University of Waterloo
- 2021–2022 Queen Elizabeth II (QEII) Graduate Scholarship - University of Waterloo
- 2017 Jack Neal Prize in Computer and Electrical Engineering - University of Alberta
- 2013-2016 Dean's Citation in Engineering - University of Alberta
- 2014 Dean's Research Award - University of Alberta

Academic Services

Conference reviewer:

- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE/RSJ International Conference on Intelligent Robotics and Systems (IROS)
- IEEE Conference on Control Technology and Applications (CCTA)

Journal reviewer:

- IEEE Transactions on Robotics (T-RO)
- IEEE Robotics and Automation Letters (RA-L)
- IEEE Transactions on Automatic Control (TAC)
- Automatica (Elsevier)

Technical Skills

- Languages Python, C++ (14/17), C, C#, Matlab, Objective-C
- Technologies ROS2, ML Libraries (PyTorch, NumPy, scikit-learn), Gymnasium, MuJoCo, Docker, Unity, MQTT, Git

Other Projects

RoVR the Explorer: VR-controlled ground robot with a point cloud stream to visualize its environment. (*3rd Place Win at HackED 2019*). [GitHub]

Paulie Blart Security Robot: Autonomous security robot to detect intruders (using CNNs). (*1st Place Win at SFHacks 2018*). [GitHub] [Featured on Global News]